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Education

SIES Graduate School of Technology, B.Tech in Artificial Intelligence And Machine Learning 2021 – 2025

Experience

Python AI Engineer at Arcitech.ai, Navi Mumbai Feb 2026 – Present

- Designed and shipped multi-agent AI systems in production using **LangChain**, **LangGraph**, and **LangSmith** for orchestration, monitoring, and latency optimization.
- Re-engineered image upscaling pipeline for native 2K/4K outputs; fixed aspect-ratio hallucination defects; drove GPT-5.4 model upgrade via comparative benchmark analysis.
- Built LLM-ready web-scraping data pipeline replacing unreliable PDFs, creating a scalable foundation for advanced RAG; proactively audited APIs and resolved production data issues.
- Designed clickable AI suggestion cards as a first-class response pattern, significantly improving user discoverability and engagement.

Python AI Intern at Arcitech.ai, Navi Mumbai Dec 2025 – Feb 2026

- Built multi-agent PR generation system (ADFactors) using **LangChain** and **LangGraph**; integrated compliance agents, web search, and PDF generation with team collaboration.
- Monitored agent performance with **LangSmith**; reduced latency by swapping high-latency reasoning models with task-optimised alternatives.
- Developed REST APIs with **FastAPI**, **PostgreSQL**, and **pgvector**; built RAG systems with persistent memory during training phase.

AI/ML Intern at Annam.ai, IIT Ropar – Center of Excellence in Digital Agriculture Aug 2025 – Oct 2025

- Conducted computer vision research using **PyTorch**, **Transformers**, and **Vision Transformers (ViT)** for high-performance agricultural image modeling.
- Applied **MLOps** with **MLflow** for experiment tracking and model versioning; deployed with **FastAPI** and **Docker**; explored LLMs for intelligent automation.

Projects

GuardNet: Violence Detection in Public Surveillance [GuardNet](#)

- Hybrid deep learning model (ConvNeXt-Tiny + LSTM + attention) for CCTV violence detection; published at [IEEE](#).
- **Tools:** PyTorch, ConvNeXt, LSTM, OpenCV, CUDA | **Keywords:** Computer Vision, Temporal Modeling, Surveillance AI

RAG for Hotels [RAG Model for Hotels](#)

- RAG pipeline for hotel booking insights using FAISS, Mistral-7B (Hugging Face), and LangChain; deployed with FastAPI and Docker.
- Performed comprehensive data cleaning and EDA to improve retrieval quality and downstream response accuracy.

Financial Fraud Detection System [Fraud Detection](#)

- ML model on 6.3M transactions (0.13% fraud); Random Forest & XGBoost with SMOTE for class imbalance handling.
- Engineered domain-specific features to improve precision and recall; evaluated models using AUC-ROC and F1 metrics.

News Article Translation & TTS in Indic Languages [News TTS](#)

- End-to-end pipeline for extraction, translation, and speech synthesis using IndicTrans2 and Facebook TTSMMMS; deployed via FastAPI and Docker.
- Supports multiple Indic languages with modular architecture, enabling easy extension to new language pairs and TTS voices.

Technologies

Languages: Python, JavaScript, SQL, Bash, HTML

Tools & Platforms: Docker, AWS, Git/GitHub, LangChain, LangGraph, LangSmith, FastAPI, Django

Databases: PostgreSQL, MySQL **Data Viz:** Matplotlib, Seaborn, Plotly, Tableau

ML Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, NLTK, SpaCy, NumPy, Pandas

AI & NLP: Transformers, GPT-4, Llama-3, Claude 3 | Generative AI, RAG, NER, Chatbots, Semantic Search